EE1007
Introduction to Computer Science Laboratory

Lab 7
Recursion

Andy, Yu-Guang Chen
Assistant Professor, Department of EE
National Central University
andyygchen@ee.ncu.edu.tw

Lab 7.1
Square Root

- In this problem, you need to write a program to calculate the square root of any positive integers
- Then, the program will display the result with tenth decimal place.
- The eleventh decimal place and the following places are unconditionally discard.

Lab 7.1
Square Root

- The program need to provide fool-proof to avoid program errors.
- **You are not allowed to use Sqrt() in C++ standard library**. If it is used, no score will be given.
- Please write the program in a **recursive manner**. If recursion is not used, no score will be given.
  - You need to define function(s) and call itself either directly or indirectly
- The range of each input integer is [0~2^{63}-1]

Lab 7.1
Square root

- The screen dialog should appear as follows:

```
Enter a number: -222
The number is out of range. Please enter again: 5
2.2360679774

Process returned 0 (0x0)   execution time : 4.376 s
Press any key to continue.
```

Lab 7.2
Combining Sequence Numbers

- In this problem, you need to write a program which can sort two given number sequences and merge them together.
- A number sequence is defined as a series of digits consists of [1-9] with arbitrary appear numbers and order. For example: 94877 is a sequence consists of 5 digits 9, 4, 8, 7, 7 where 7 appears twice. Note that 0 will not be appeared in the sequence.
- In the beginning, the program will take two number sequence as inputs, and sort each number sequence in **increasing order**.
- Then, the program will merge two sequence numbers in **decreasing order**.

Lab 7.2
Combining Sequence Numbers

- The range of each input number sequence is [1~2^{63}-1].
- In this question, we assume no out-of-range value will be given, so you do not need to consider fool-proof.
- Please use the sample code LAB7_2_sample.cpp that we provide on ee-class as a template to complete your code.
- Please write the program in a **recursive manner** and you cannot using any **while, do… while, and for** loop.
- If recursion is not used or using any **while, do… while, and for loop**, no score will be given.

Lab 7.2
Combining Sequence Numbers

- The screen dialog should appear as follows:

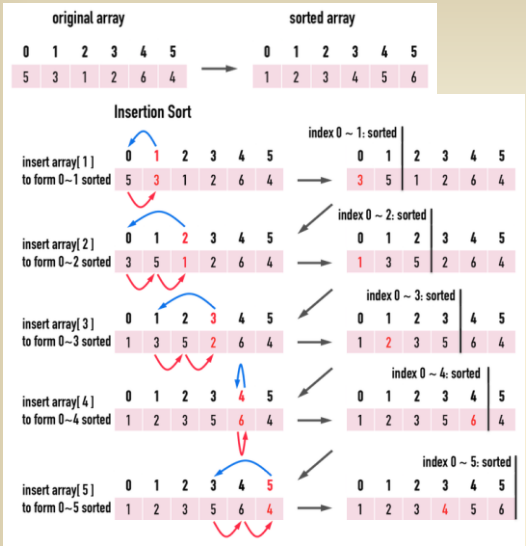
```
Please enter two sequence numbers:
147258369
741852963
Sorted result of the first sequence number: 123456789
Sorted result of the second sequence number: 123456789
Result: 998877665544332211

Process returned 0 (0x0)   execution time : 5.785 s
Press any key to continue.
```


Lab 7.2 Hint: Insertion Sort

- Insertion sort is used to construct ordered sequences.
- For unsorted data, scan from back to front in the sorted sequence, find the corresponding position and insert it.
- Algorithm:
  1. Starting from the first element, the element can be considered to have been sorted.
  2. Take out the next element and scan from back to front in the sorted sequence of elements.
  3. If the element (sorted) is greater than the new element, move the element to the next position.
  4. Repeat step 3 until you find the position where the sorted element is less than or equal to the new element.
  5. After inserting the new element at that position.
  6. Repeat steps 2~5.

Lab 7.2 Hint: Insertion Sort



- Refernece：
- Insertion Sort
- http://alrightchiu.github.io/SecondRound/comparison-sort-insertion-sortcha-ru-pai-xu-fa.html

Lab 7.3
Multi-Function Calculator

- In this problem, you are asked to complete a multi-function calculator which can perform addition, subtraction, and factorization on a local variable ***number***. Partial code of this problem is given in LAB7_3_sample.cpp

```
#include <iostream>

using namespace std;


int number_of_addition=1,number_of_subtracttion=1;

void add(){

}
void sub(){

}
void factorization(){

}

int main()
{
    int number=0,choose;
    while(true){
        cout <<"The number now is "<<number<<endl;
        cout << "Choose a function:" << endl;
        cout << "1. Add    2. Sub    3. Factorization    4. Exit"<<endl<<endl;;
        cin>>choose;
    }

    return 0;
}
```

Lab 7.3
Multi-Function Calculator



- You must complete the following functions:
  - ***void add(…)***
    - When the add function is called, it will firstly update the local variable "number" by adding the value of "number" and "number_of_addition", and then increase the value of global variable "number_of_addition" by one.
  - ***void sub(…)***
    - When the sub function is called, it will firstly update the local variable "number" by subtract the value "number_of_subtraction" from "number" (i.e., number- number_of_subtraction), and then increase the value of global variable "number_of_subtraction" by one.
  - ***void factorization(…)***
    - When the factorization function is called, it will print all factorizations of the variable "number" (included) in an increasing order with comma-separated sequence.



Lab 7.3
Multi-Function Calculator



- You CANNOT declare any other **global variable** in this program and any other **local variable** in ***main(), add(…)*** and ***sub(…)*** function. You also CANNOT change the return types of the given functions.
- The program need fool-proof when the user entered undefined action or it is not possible to display the factorization.



Lab 7.3
Multi-Function Calculator

◆The screen dialog should appear as follows:

```
The number now is 0
Choose a function:
1. Add    2. Sub    3. Factorization    4. Exit

1

The number now is 1
Choose a function:
1. Add    2. Sub    3. Factorization    4. Exit

1

The number now is 3
Choose a function:
1. Add    2. Sub    3. Factorization    4. Exit

2

The number now is 2
Choose a function:
1. Add    2. Sub    3. Factorization    4. Exit

3
2 = 1,2

The number now is 2
Choose a function:
1. Add    2. Sub    3. Factorization    4. Exit

4
```

```
The number now is 0
Choose a function:
1. Add    2. Sub    3. Factorization    4. Exit

5
Enter error!!

The number now is 0
Choose a function:
1. Add    2. Sub    3. Factorization    4. Exit

2

The number now is -1
Choose a function:
1. Add    2. Sub    3. Factorization    4. Exit

3
No factorization

The number now is -1
Choose a function:
1. Add    2. Sub    3. Factorization    4. Exit

4
```


Evaluation

◆Lab7.1(30%)
- ➢5 testcases, 6% each

◆Lab7.2(40%)
- ➢5 testcases, 8% each

◆Lab7.3(30%)
- ➢5 testcases, 6% each





Andy, Yu-Guang Chen

Assistant Professor, Department of EE, NCU

Email: andyygchen@ee.ncu.edu.tw